RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

Academic Year 2018-2019 (Even Semester)

<u>Innovative practice(s) description</u>

Course code and title: ME8492 Kinematics of Machinery

Name of the Instructor: Mr.L.Karthikeyan, AP/Mechanical

Dr.S.Godwin Barnabas, AP(SG)/Mechanical

Date & Time: 01.03.2019 & 4.10 pm

T2P (Theory to Practical) *Basic Kinematic Concepts of simple mechanisms* and *'Gears and Gear Trains'*

- After completion of the UNIT II: Kinematics of linkage mechanism & unit IV Gears and gear trains in ME8492 Kinematics of Machinery, it is expected that the student will be able to understand and calculate the velocity and acceleration for the various mechanisms like four bar mechanism, single slider crank chain mechanism etc.

- Students will be able to visualize and understand the various gear terminologies and arrangements in simple, compound and epicyclical gear trains.

Through this activity, the students are made to see lively the kinematics of simple mechanisms and the gear profile terms and its gear ratio.

OUTCOME:

- This theory to practice was important to attain practical hands-on experience.
- Student will understand the basics of various linkage mechanisms.
- Students will explain the velocity and acceleration concepts for of various linkage mechanisms.
- Students will be able to explain the gear types and its related terminologies

This activity helps in attaining the following CO – PO mapping:

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO9**	**PO12**
CO2: The student will be able to calculate velocity and acceleration of simple Mechanisms	2	3	2	3	2	2	1
CO4: The student will be able to solve problems on gears and gear trains	3	3	2	2	2	2	1